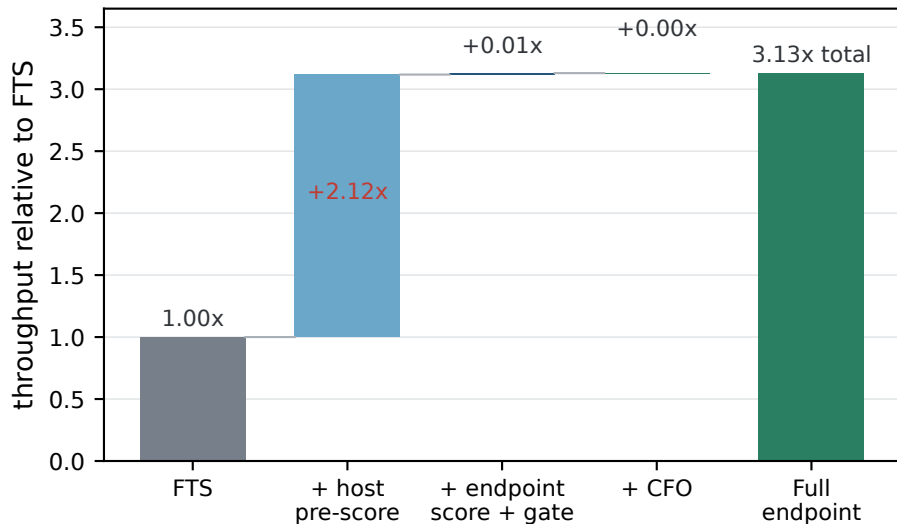
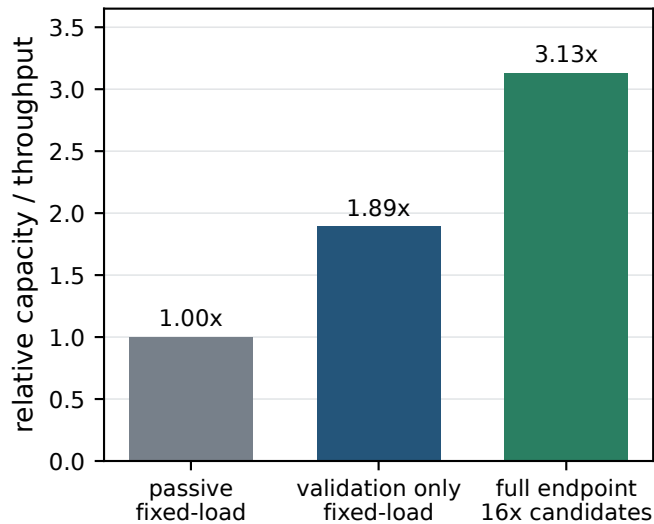


Scope of the 3.1x claim: validation protects atomicity; pre-transfer filtering drives throughput

(a) Same-point cumulative throughput decomposition



(b) Keep validation-only and headline scopes separate



Runtime validation-only capacity comes from the measured stale-event replay. Waterfall stages are SimCXL results at one identical point (2 GB/s, 16 producers, 16x candidates). CFO adds no throughput here because the endpoint has already reached the compute ceiling.